

Assignments based on NumPy

1) NumPy Basics

1. Create a NumPy array of integers from 1 to 20.
2. Find the shape, size, and data type of the array.
3. Convert the array into float type.
4. Find the sum and mean of the array.

2) Array Indexing & Slicing

1. Create an array of 15 random integers between 10 and 50.
2. Extract:
 - First 5 elements
 - Last 5 elements
3. Replace all values greater than 30 with 0.
4. Print elements at even indices.

3) 2D Arrays & Matrix Operations

1. Create a 3×3 matrix with values from 1 to 9.
2. Find the transpose of the matrix.
3. Calculate:
 - Row-wise sum
 - Column-wise sum
4. Find the determinant of the matrix.

4) Random Numbers & Statistics

1. Generate 100 random numbers between 0 and 1.
2. Find:
 - Mean
 - Median
 - Standard deviation
3. Count how many values are greater than 0.5.
4. Normalize the data between 0 and 1.

5) Mathematical Operations

1. Create two arrays of size 10.
2. Perform:
 - Addition
 - Subtraction
 - Multiplication
 - Division
3. Find:
 - Square root
 - Square
 - Exponential of elements

6) Conditional Operations

1. Create an array of 20 random integers between 1 and 100.
2. Extract all even numbers.
3. Replace odd numbers with -1.
4. Count numbers divisible by 5.

7) Reshaping & Stacking

1. Create a 1D array of size 12.
2. Reshape it into:
 - 3×4
 - 2×6
3. Stack two arrays vertically and horizontally.
4. Flatten the final array.

8) Sorting & Searching

1. Create an array of 10 random integers.
2. Sort the array in:
 - Ascending order
 - Descending order
3. Find the maximum and minimum values.
4. Find the index of the maximum value.

9) Linear Algebra

1. Create two 3×3 matrices.
2. Perform:
 - Matrix multiplication
 - Element-wise multiplication
3. Find:
 - Inverse of a matrix
 - Eigenvalues

10) Mini Data Analysis

1. Generate a dataset of 50 students with marks in 5 subjects.
2. Calculate:
 - Total marks
 - Average marks
3. Find:
 - Topper student index
 - Number of students scoring above 75%
4. Normalize the dataset.